

Power BI Work Documentation

Project Title

Student Knowledge Hub Analytics Dashboard

Tool Used

Microsoft Power BI Desktop

Dataset Used

Open University Learning Analytics Dataset (OULAD)

The dataset was downloaded as a ZIP file and extracted. It contains multiple CSV files related to students, courses, learning resources, and student interactions with an online learning environment.

Main Goal of This Part

The goal of this Power BI part is to analyze student learning activity and show how a Student Knowledge Hub can use analytics to understand:

- Student engagement
- Course activity
- Learning resource usage
- Student final results
- Interactive filtering by course and result status

1. Dataset Files Imported into Power BI

The following CSV files were imported into Power BI Desktop:

File Name	Purpose
studentInfo.csv	Contains student information such as student ID, gender, region, studied credits, and final result.
courses.csv	Contains course/module information and presentation length.
vle.csv	Contains information about learning resources and activity types.
studentVle.csv	Contains student interactions/clicks with learning resources.

These files were imported using:

Home → Get Data → Text/CSV → Load

2. Data Model Preparation

2.1 Creating CourseKey

Since `code_module` alone was not enough to uniquely identify a course presentation, a new calculated column called `CourseKey` was created.

The key combines:

- `code_module`
- `code_presentation`

This creates values such as:

- `AAA-2013J`
- `BBB-2014B`
- `FFF-2014J`

DAX formulas used

In `courses` table

```
CourseKey = courses[code_module] & "-" & courses[code_presentation]
```

In `studentInfo` table

```
CourseKey = studentInfo[code_module] & "-" & studentInfo[code_presentation]
```

In `studentVle` table

```
CourseKey = studentVle[code_module] & "-" & studentVle[code_presentation]
```

In `vle` table

```
CourseKey = vle[code_module] & "-" & vle[code_presentation]
```

3. Relationships Created

The following relationships were created in the Model View:

From Table	Column	To Table	Column	Relationship Type
courses	CourseKey	studentInfo	CourseKey	One-to-Many
courses	CourseKey	vle	CourseKey	One-to-Many
vle	id_site	studentVle	id_site	Many-to-One / automatically detected

A direct relationship between `studentInfo` and `studentVle` was not used because Power BI showed an ambiguous path warning. To avoid model confusion, that relationship was cancelled.

4. Dashboard Page Created

A main dashboard page was created with the title:

Student Knowledge Hub Analytics Dashboard

The dashboard includes KPI cards, charts, and slicers.

5. KPI Cards Created

5.1 Total Students

Visual used: Card

Field used:

- Table: `studentInfo`
- Field: `id_student`
- Aggregation: Count distinct

Result: Approximately `28.785K`

This card shows the number of unique students in the dataset.

5.2 Total Interactions

Visual used: Card

Field used:

- Table: `studentVle`
- Field: `sum_click`

- Aggregation: Sum

Result: Approximately 40M

This card shows the total number of student clicks/interactions with learning resources.

5.3 Total Courses

Visual used: Card

Field used:

- Table: courses
- Field: CourseKey
- Aggregation: Count distinct

Result: 22

This card shows the number of unique course presentations.

5.4 Total Learning Resources

Visual used: Card

Field used:

- Table: vle
- Field: id_site
- Aggregation: Count distinct

Result: Approximately 6.34K

This card shows the number of unique learning resources in the dataset.

6. Charts Created

6.1 Interactions by Course

Visual used: Clustered bar chart

Fields used:

- Y-axis: studentVle[CourseKey]
- X-axis: studentVle[sum_click]
- Aggregation: Sum

This chart shows which courses have the highest student interaction levels.

The chart was sorted in descending order, so the course with the highest number of clicks appears first.

6.2 Student Final Results

Visual used: Donut chart

Fields used:

- Legend: `studentInfo[final_result]`
- Values: `studentInfo[id_student]`
- Aggregation: Count distinct

This chart shows the distribution of student final results:

- Pass
 - Fail
 - Withdrawn
 - Distinction
-

6.3 Resource Type Usage

Visual used: Bar chart

Fields used:

- Y-axis: `vle[activity_type]`
- X-axis: `studentVle[sum_click]`
- Aggregation: Sum

This chart shows which learning resource types receive the most student clicks.

Examples of activity types include:

- oucontent
 - forumng
 - quiz
 - homepage
 - subpage
 - resource
-

7. Slicers Added

7.1 Course Filter

Visual used: Slicer

Field used:

- Table: `courses`
- Field: `CourseKey`

This slicer allows users to filter the dashboard by a specific course presentation.

Example selected value:

`BBB-2013B`

When this filter was tested, the dashboard changed correctly and showed values only for the selected course.

7.2 Final Result Filter

Visual used: Slicer

Field used:

- Table: `studentInfo`
- Field: `final_result`

This slicer allows users to filter the dashboard by student outcome:

- Pass
- Fail
- Withdrawn
- Distinction

The slicer was tested and the dashboard responded correctly.

8. Dashboard Layout

The dashboard was arranged as follows:

1. Title at the top.
2. Four KPI cards below the title.
3. Main charts under the KPI cards.
4. Slicers placed on the side for filtering.

The final layout includes:

- Total Students
 - Total Interactions
 - Total Courses
 - Total Learning Resources
 - Interactions by Course
 - Student Final Results
 - Resource Type Usage
 - CourseKey filter
 - Final Result filter
-

9. File Saved

The Power BI file was saved as:

Student_Knowledge_Hub_Dashboard.pbix

This file contains the imported dataset, relationships, calculated columns, dashboard visuals, and slicers.

10. Explanation for Report

This Power BI dashboard supports the Student Knowledge Hub project by transforming raw learning analytics data into useful knowledge. The dashboard helps decision-makers understand how students interact with learning resources, which courses are most active, what types of resources are used most, and how student outcomes are distributed.

By connecting Power BI with the knowledge management idea, the system does not only store learning materials; it also analyzes usage patterns and supports better decisions about course content, student support, and resource improvement.

11. Screenshots to Include Later

The following screenshots should be added to the final documentation:

1. Dataset files imported in Power BI.
 2. Model View showing relationships between tables.
 3. Dashboard overview.
 4. Dashboard filtered by CourseKey.
 5. Dashboard filtered by final_result.
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12. Current Status

Power BI dashboard part is completed at a basic functional level.

Completed:

- Dataset imported
- Keys created
- Relationships created
- KPI cards created
- Charts created
- Slicers created
- Dashboard tested
- Power BI file saved

Next part:

SharePoint setup for the Student Knowledge Hub system.